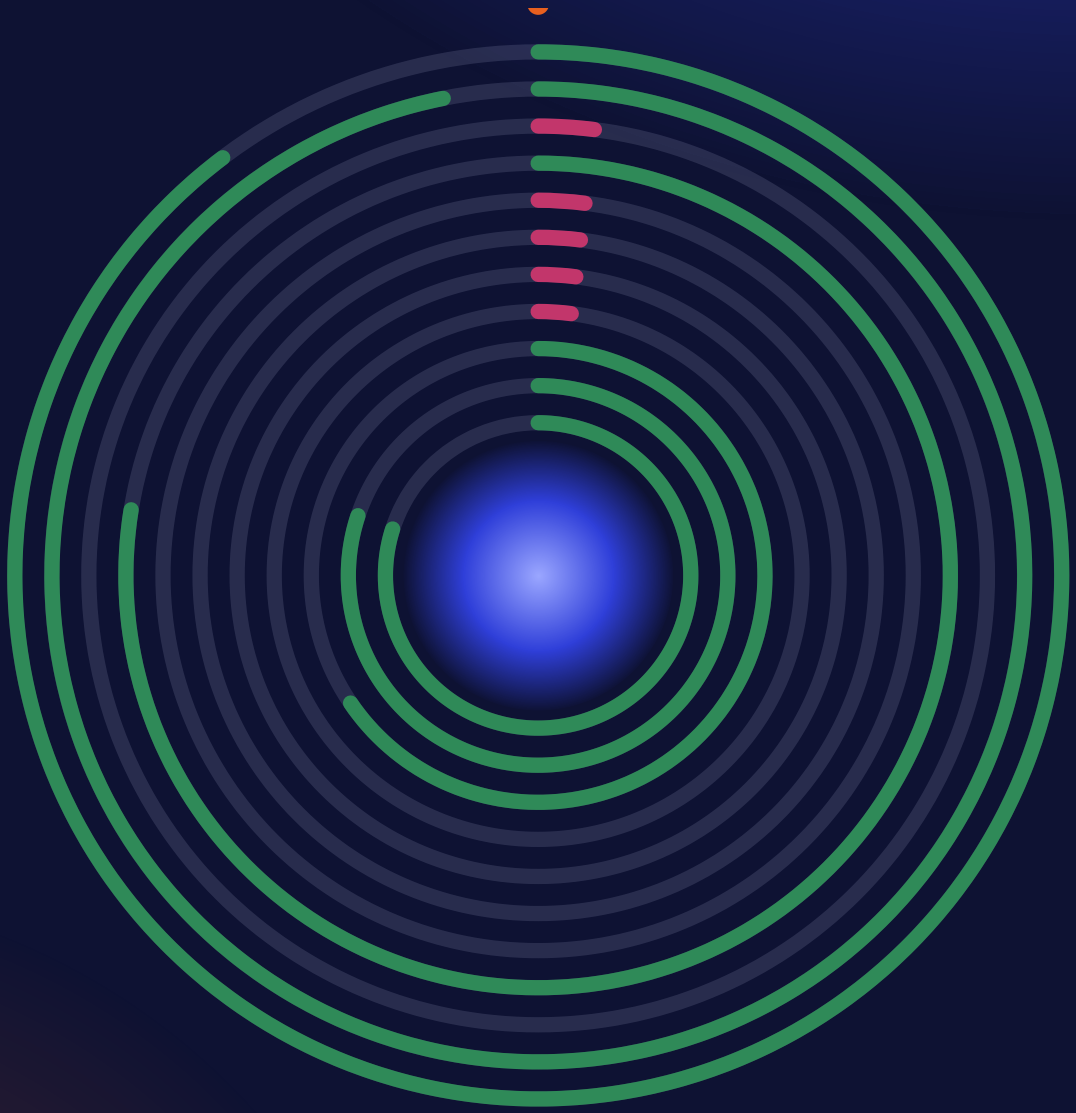
Assay

LLM Release Claims Audit

Do vendor claims survive independent measurement?



Prepared by Bhavin Maheshwari, Group CTO, PlanetAI

13 Nov 2026 · Scheduled run

Each ring is one claim or stress test; arc length shows how much of it held

About this report

What we checked, against what evidence, and how to read the verdicts.

Scope

This edition reflects the Assay scheduled run of 13 Nov 2026. That run detected 1 new model (gemini-4-hydrogen). It is a desk audit: every verdict rests on published vendor and independent measurements, cited in each chapter; no live API tests were run. It covers 24 model releases announced between 24 Nov 2025 and 13 Nov 2026: 15 vendor claims, 6 Assay stress tests and 2 test hypothesises. Vendor claims are paraphrased from each vendor's announcement; stress tests re-check a claim outside the conditions the vendor stated, and are never counted against the vendor.

Evidence base

50 sources: 27 vendor announcements and pricing pages, 14 independent measurements, and 9 secondary reports used where no primary figure was available. Every number carries a numbered citation.

Reuse control

A verdict records what was true under the conditions it was measured in. Whether it may still inform a new decision is checked separately on every run: each verdict carries a fingerprint of its model versions, settings, prices and evidence, and is cleared for reuse only while that fingerprint is unchanged, the evidence is under 90 days old and no newer model version has appeared. In this edition 8 of 15 verdicts are cleared for reuse.

Verdict key

How much of the claimed improvement independent data confirms

Hallmarked	Plated	Fool's gold	Unassayed
verified	partly true	marketing	no public data
Delivered at least 80% of the claim.	Delivered 40–79%, or delivered in full but measured quality fell.	Under 40% of the claim, or it got worse.	No independent measurement yet. Run the live pack.

Solid badges are vendor claims. Outlined badges are Assay stress tests (marked Held or Reversed) and hypotheses; they are never counted against the vendor.

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16. Google: gemini-1.5-flash-exp against gemini-2.5-flash-exp (draft)
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Executive summary

6 of 8 vendor claims held up on independent data; 1 held only in part and 1 fell short.

All 6 savings we re-tested beyond the vendors' stated settings reversed: the gains are real, but only where the vendor said they would be.

15
vendor claims examined across 24 releases

75%
of vendor claims with independent data held up in full

6 of 6
stress tests where the saving reversed outside the stated conditions

Exhibit 1

Verdicts on vendor claims



Vendor claims only; stress tests and hypotheses are reported separately. Source: Assay analysis of vendor announcements against independent measurements, cited in each chapter.

Key findings

- **Google, Gemini 4 Argon (introductory price): claim held.** Leads rival frontier models on the Vals Index, ahead of Opus 5.5. Claimed 1.9 points higher; independent data shows 1.9 points higher.[21, 22]
- **Anthropic, Opus 5.5 (medium): claim held.** Costs about 40% less to run than Opus 5 on typical workloads at default effort. Claimed 40% cheaper; independent data shows 39% cheaper.[33, 34]
- **OpenAI, GPT-6 Luna: held in part.** GPT-6 Luna cuts prices by about half. Claimed 50% cheaper; independent data shows 61% cheaper, with lower measured quality.[40]
- **Google, Gemini 3.8 Flash: fell short.** Completed more than three times as many tasks as 3.7 Flash. Claimed 200% higher score; independent data shows 11% higher score.[46]
- **Settings matter.** The same saving, after the introductory price ends: 18% more expensive. This is outside what the vendor claimed, but it is where an unpinned configuration can land.[23, 22]
- **7 vendor claims have no independent data yet.** Live test packs are ready so they can be measured on your own workload.

What leaders should do

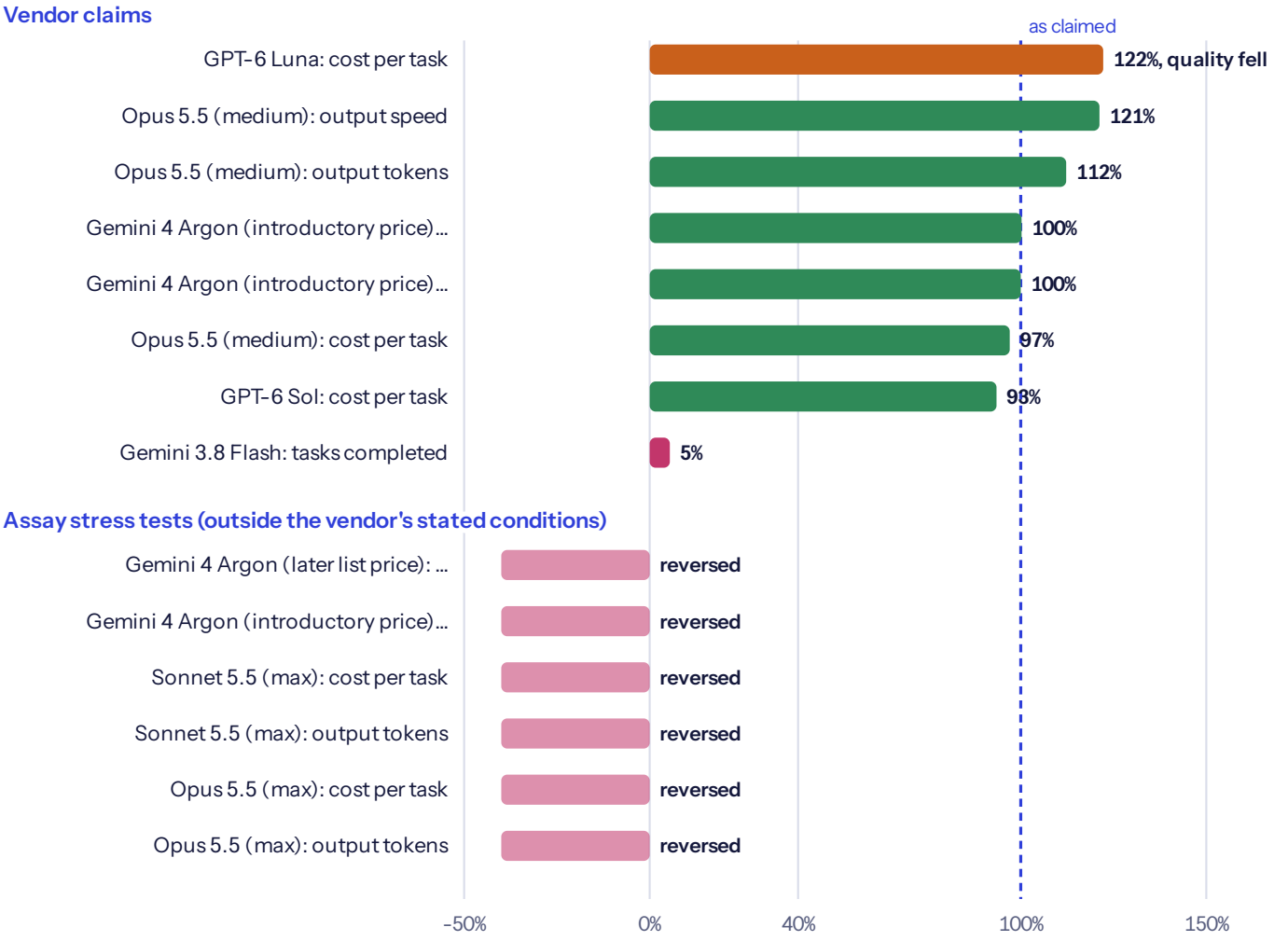
Test at the effort level you will actually deploy, budget on cost per correct answer rather than price per token, treat any single-benchmark claim as a hypothesis until it survives a noise check, and re-run the audit on every release.

The evidence at a glance

Savings hold at the settings vendors quote; push the settings further and some reverse.

Exhibit 2

Share of each claimed improvement confirmed by independent data



Bars are coloured by verdict; stress tests are shaded lighter. "Quality fell" marks a saving that came with a lower benchmark score.
Source: independent measurements cited in each chapter; Assay analysis.

Exhibit 3

Verdict matrix by claim type

Release	Cost	Speed	Tokens	Quality
Google gemini-4-hydrogen against gemini-4-carbon	–	–	–	–
Google gemini-4-carbon against gemini-3.7-flash	–	–	–	–
Anthropic claude-fable-5-2 against claude-fable-5-1	–	–	–	–
Anthropic claude-mythos-5-2 against claude-mythos-5-1	–	–	–	–
OpenAI gpt-6-cyber against gpt-5.6-cyber	–	–	–	–
Google gemini-2.5-pro-models against gemini-2.5-pro-preview	–	–	–	–
Google gemini-2.5-flash-models against gemini-2.5-flash-native	–	–	–	–
Anthropic claude-opus-4-8 against claude-opus-4-5-20251101	–	–	–	–
OpenAI gpt-6-rosalind against gpt-6-astra	–	–	–	–
OpenAI gpt-6.1-terra against gpt-5.6-terra	–	–	–	–
Anthropic claude-tag-2026-10 against claude-sonnet-4-5	–	–	–	–
Google gemini-1.5-pro-exp against gemini-2.5-pro-exp	–	–	–	–
Google gemini-1.5-flash-exp against gemini-2.5-flash-exp	–	–	–	–

Release	Cost	Speed	Tokens	Quality
Anthropic claude-haiku-5-5 against claude-opus-5-5	–	–	–	–
Google gemini-2.5-pro-pre against gemini-2.5-pro-preview	–	–	–	–
Anthropic claude-sonnet-4-5-20250929 against claude-sonnet-4-5	–	–	–	–
Google gemini-1.5-flash-latest against gemini-1.5-flash	–	–	–	–
Google gemini-1.5-pro-latest against gemini-1.5-pro	–	–	–	–
Google Gemini 4 Argon against Claude Opus 5.5	Hallmarked Reversed Reversed	–	–	Hallmarked Hallmarked Unassayed
Anthropic Claude Sonnet 5.5 against Sonnet 5	Unassayed Reversed	Unassayed	Unassayed Reversed	–
Anthropic Claude Opus 5.5 against Opus 5	Hallmarked Reversed	Hallmarked	Hallmarked Reversed	–
OpenAI GPT-6 Sol and Luna against GPT-5.6	Hallmarked Plated	–	–	Unassayed
Google Gemini 3.8 Flash against 3.7 Flash	–	–	Unassayed	Fool's gold
Anthropic Claude Opus 4.5 token efficiency against Sonnet 4.5	Unassayed	–	Unassayed	–

Outlined badges are Assay stress tests or hypotheses, not vendor claims.

Market context

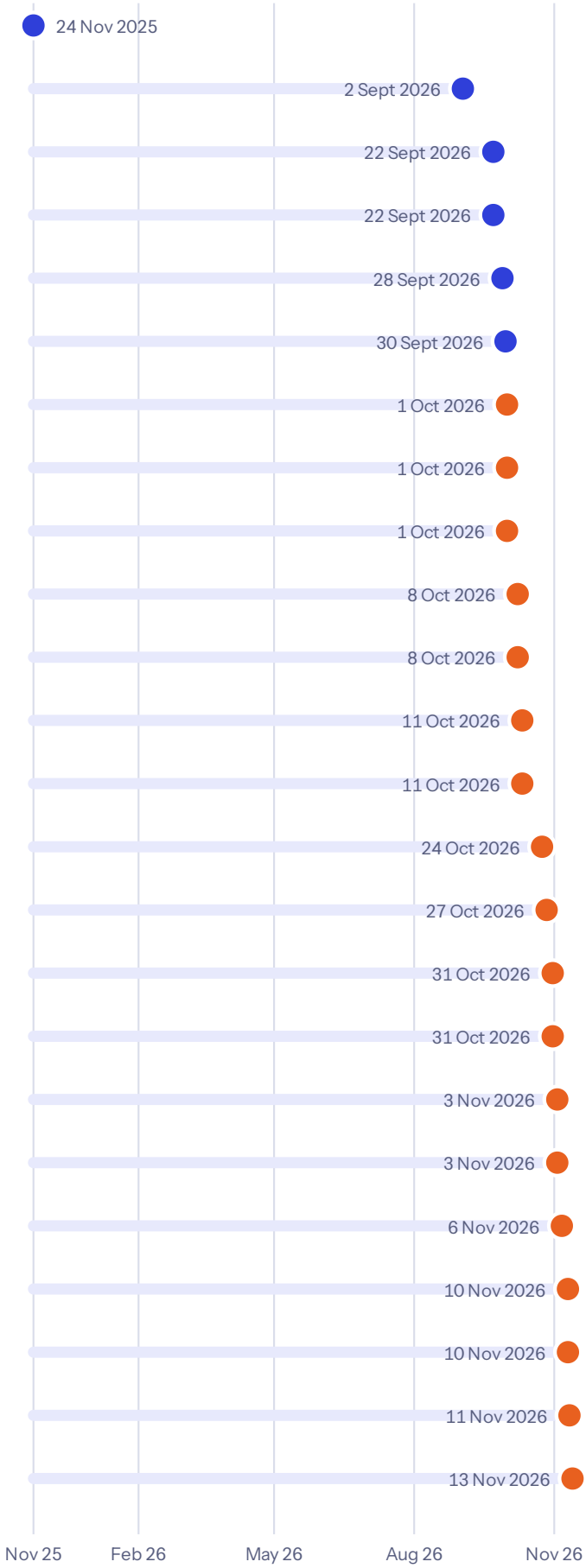
3 of 5 upgrades cut list prices, by 20% to 58% on

output; 2 held their price.

Exhibit 4

Release timeline

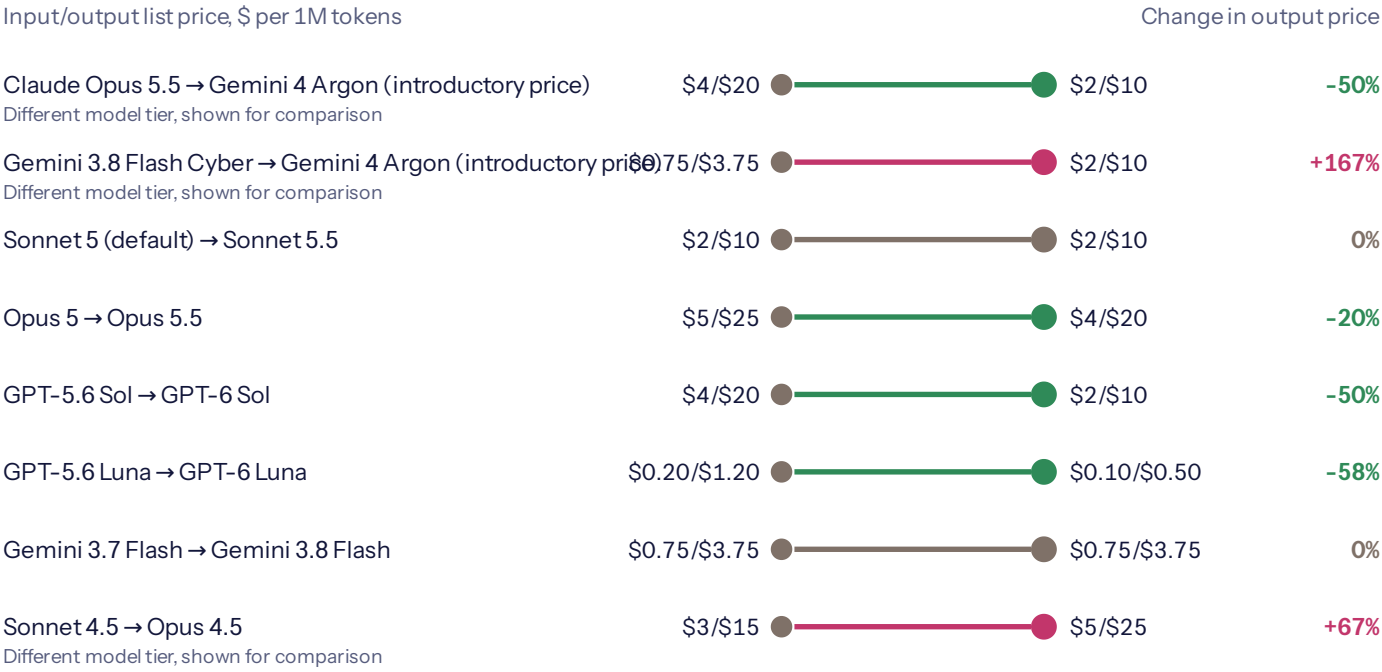
Anthropic	Claude Opus 4.5 token efficiency agai…	24 Nov 2025
Google	Gemini 3.8 Flash against 3.7 Flash	2 Sept 2026
Anthropic	Claude Opus 5.5 against Opus 5	22 Sept 2026
OpenAI	GPT-6 Sol and Luna against GPT-5.6	22 Sept 2026
Anthropic	Claude Sonnet 5.5 against Sonnet 5	28 Sept 2026
Google	Gemini 4 Argon against Claude Opus 5.5	30 Sept 2026
Anthropic	claude-sonnet-4-5-20250929 against cl…	1 Oct 2026
Google	gemini-1.5-flash-latest against gemin…	1 Oct 2026
Google	gemini-1.5-pro-latest against gemini-…	1 Oct 2026
Anthropic	claude-haiku-5-5 against claude-opus-…	8 Oct 2026
Google	gemini-2.5-pro-pre against gemini-2.5…	8 Oct 2026
Google	gemini-1.5-pro-exp against gemini-2.5…	11 Oct 2026
Google	gemini-1.5-flash-exp against gemini-2…	11 Oct 2026
Anthropic	claude-tag-2026-10 against claude-son…	24 Oct 2026
OpenAI	gpt-6.1-terra against gpt-5.6-terra	27 Oct 2026
Anthropic	claude-opus-4-8 against claude-opus-4…	31 Oct 2026
OpenAI	gpt-6-rosalind against gpt-6-astra	31 Oct 2026
Google	gemini-2.5-pro-models against gemini-…	3 Nov 2026
Google	gemini-2.5-flash-models against gemin…	3 Nov 2026
OpenAI	gpt-6-cyber against gpt-5.6-cyber	6 Nov 2026
Anthropic	claude-fable-5-2 against claude-fable…	10 Nov 2026
Anthropic	claude-mythos-5-2 against claude-myth…	10 Nov 2026
Google	gemini-4-carbon against gemini-3.7-fl…	11 Nov 2026
Google	gemini-4-hydrogen against gemini-4-ca…	13 Nov 2026



Blue: reviewed scenarios. Orange: drafts awaiting review.

Exhibit 5

List price per release



List prices at release, from the vendor pricing pages and announcements cited in each chapter. A lower price per token lowers cost per task only if token use does not rise.

Assay | LLM Release Claims Audit | 13 Nov 2026

Bhavin Maheshwari, Group CTO

Chapter 1: Google (draft, unreviewed)

gemini-4-hydrogen against gemini-4-carbon

Released 13 Nov 2026. Sources checked 13 Nov 2026.

In brief

Draft created by Release watch. Claims have not been checked against independent data yet. Review each claim against the source before publishing.

Claims examined

- None

Models and list prices

Model	Input	Output
gemini-4-carbon gemini-4-carbon	–	–
gemini-4-hydrogen gemini-4-hydrogen	–	–

$ per 1M tokens.

So what

There is no independent evidence yet. Run the live pack before relying on these numbers.

Exhibit 6

What independent data shows

Chapter 2: Google (draft, unreviewed)

gemini-4-carbon against gemini-3.7-flash

Released 11 Nov 2026. Sources checked 11 Nov 2026.

In brief

Draft created by Release watch. Claims have not been checked against independent data yet. Review each claim against the source before publishing.

Claims examined

- None

Models and list prices

Model	Input	Output
gemini-3.7-flash gemini-3.7-flash	–	–
gemini-4-carbon gemini-4-carbon	–	–

$ per 1M tokens.

So what

There is no independent evidence yet. Run the live pack before relying on these numbers.

Exhibit 7

What independent data shows

claude-fable-5-2 against claude-fable-5-1

Released 10 Nov 2026. Sources checked 10 Nov 2026.

In brief

Draft created by Release watch. Claims have not been checked against independent data yet. Review each claim against the source before publishing.

Claims examined

- None

Models and list prices

Model	Input	Output
claude-fable-5-1 claude-fable-5-1	–	–
claude-fable-5-2 claude-fable-5-2	–	–

$ per 1M tokens.

So what

There is no independent evidence yet. Run the live pack before relying on these numbers.

Exhibit 8

What independent data shows

claude-mythos-5-2 against claude-mythos-5-1

Released 10 Nov 2026. Sources checked 10 Nov 2026.

In brief

Draft created by Release watch. Claims have not been checked against independent data yet. Review each claim against the source before publishing.

Claims examined

- None

Models and list prices

Model	Input	Output
claude-mythos-5-1 claude-mythos-5-1	–	–
claude-mythos-5-2 claude-mythos-5-2	–	–

$ per 1M tokens.

So what

There is no independent evidence yet. Run the live pack before relying on these numbers.

Exhibit 9

What independent data shows

Chapter 5: OpenAI (draft, unreviewed)

gpt-6-cyber against gpt-5.6-cyber

Released 6 Nov 2026. Sources checked 6 Nov 2026.

In brief

Draft created by Release watch. Claims have not been checked against independent data yet. Review each claim against the source before publishing.

Claims examined

- None

Models and list prices

Model	Input	Output
gpt-5.6-cyber gpt-5.6-cyber	–	–
gpt-6-cyber gpt-6-cyber	–	–

$ per 1M tokens.

So what

There is no independent evidence yet. Run the live pack before relying on these numbers.

Exhibit 10

What independent data shows

gemini-2.5-pro-models against gemini-2.5-pro-preview

Released 3 Nov 2026. Sources checked 3 Nov 2026.

In brief

Draft created by Release watch. Claims have not been checked against independent data yet. Review each claim against the source before publishing.

Claims examined

- None

Models and list prices

Model	Input	Output
gemini-2.5-pro-preview gemini-2.5-pro-preview	–	–
gemini-2.5-pro-models gemini-2.5-pro-models	–	–

$ per 1M tokens.

So what

There is no independent evidence yet. Run the live pack before relying on these numbers.

Exhibit 11

What independent data shows

gemini-2.5-flash-models against gemini-2.5-flash-native

Released 3 Nov 2026. Sources checked 3 Nov 2026.

In brief

Draft created by Release watch. Claims have not been checked against independent data yet. Review each claim against the source before publishing.

Claims examined

- None

Models and list prices

Model	Input	Output
gemini-2.5-flash-native gemini-2.5-flash-native	–	–
gemini-2.5-flash-models gemini-2.5-flash-models	–	–

$ per 1M tokens.

So what

There is no independent evidence yet. Run the live pack before relying on these numbers.

Exhibit 12

What independent data shows

claude-opus-4-8 against claude-opus-4-5-20251101

Released 31 Oct 2026. Sources checked 31 Oct 2026.

In brief

Draft created by Release watch. Claims have not been checked against independent data yet. Review each claim against the source before publishing.

Claims examined

- None

Models and list prices

Model	Input	Output
claude-opus-4-5-20251101 claude-opus-4-5-20251101	–	–
claude-opus-4-8 claude-opus-4-8	–	–

$ per 1M tokens.

So what

There is no independent evidence yet. Run the live pack before relying on these numbers.

Exhibit 13

What independent data shows

Chapter 9: OpenAI (draft, unreviewed)

gpt-6-rosalind against gpt-6-astra

Released 31 Oct 2026. Sources checked 31 Oct 2026.

In brief

Draft created by Release watch. Claims have not been checked against independent data yet. Review each claim against the source before publishing.

Claims examined

- None

Models and list prices

Model	Input	Output
gpt-6-astra gpt-6-astra	–	–
gpt-6-rosalind gpt-6-rosalind	–	–

$ per 1M tokens.

> **So what**
> There is no independent evidence yet. Run the live pack before relying on these numbers.

Exhibit 14

What independent data shows

gpt-6.1-terra against gpt-5.6-terra

Released 27 Oct 2026. Sources checked 27 Oct 2026.

In brief

Draft created by Release watch. Claims have not been checked against independent data yet. Review each claim against the source before publishing.

Claims examined

- None

Models and list prices

Model	Input	Output
gpt-5.6-terra gpt-5.6-terra	–	–
gpt-6.1-terra gpt-6.1-terra	–	–

$ per 1M tokens.

So what

There is no independent evidence yet. Run the live pack before relying on these numbers.

Exhibit 15

What independent data shows

claude-tag-2026-10 against claude-sonnet-4-5

Released 24 Oct 2026. Sources checked 24 Oct 2026.

In brief

Draft created by Release watch. Claims have not been checked against independent data yet. Review each claim against the source before publishing.

Claims examined

- None

Models and list prices

Model	Input	Output
claude-sonnet-4-5 claude-sonnet-4-5	–	–
claude-tag-2026-10 claude-tag-2026-10	–	–

$ per 1M tokens.

So what

There is no independent evidence yet. Run the live pack before relying on these numbers.

Exhibit 16

What independent data shows

gemini-1.5-pro-exp against gemini-2.5-pro-exp

Released 11 Oct 2026. Sources checked 11 Oct 2026.

In brief

Draft created by Release watch. Claims have not been checked against independent data yet. Review each claim against the source before publishing.

Claims examined

- None

Models and list prices

Model	Input	Output
gemini-2.5-pro-exp gemini-2.5-pro-exp	–	–
gemini-1.5-pro-exp gemini-1.5-pro-exp	–	–

$ per 1M tokens.

So what

There is no independent evidence yet. Run the live pack before relying on these numbers.

Exhibit 17

What independent data shows

gemini-1.5-flash-exp against gemini-2.5-flash-exp

Released 11 Oct 2026. Sources checked 11 Oct 2026.

In brief

Draft created by Release watch. Claims have not been checked against independent data yet. Review each claim against the source before publishing.

Claims examined

- None

Models and list prices

Model	Input	Output
gemini-2.5-flash-exp gemini-2.5-flash-exp	–	–
gemini-1.5-flash-exp gemini-1.5-flash-exp	–	–

$ per 1M tokens.

So what

There is no independent evidence yet. Run the live pack before relying on these numbers.

Exhibit 18

What independent data shows

claude-haiku-5-5 against claude-opus-5-5

Released 8 Oct 2026. Sources checked 8 Oct 2026.

In brief

Draft created by Release watch. Claims have not been checked against independent data yet. Review each claim against the source before publishing.

Claims examined

- None

Models and list prices

Model	Input	Output
claude-opus-5-5 claude-opus-5-5	$4.00	$20.00
claude-haiku-5-5 claude-haiku-5-5	–	–

$ per 1M tokens.

So what

There is no independent evidence yet. Run the live pack before relying on these numbers.

Exhibit 19

What independent data shows

gemini-2.5-pro-pre against gemini-2.5-pro-preview

Released 8 Oct 2026. Sources checked 8 Oct 2026.

In brief

Draft created by Release watch. Claims have not been checked against independent data yet. Review each claim against the source before publishing.

Claims examined

- None

Models and list prices

Model	Input	Output
gemini-2.5-pro-preview gemini-2.5-pro-preview	–	–
gemini-2.5-pro-pre gemini-2.5-pro-pre	–	–

$ per 1M tokens.

So what

There is no independent evidence yet. Run the live pack before relying on these numbers.

Exhibit 20

What independent data shows

claude-sonnet-4-5-20250929 against claude-sonnet-4-5

Released 1 Oct 2026. Sources checked 1 Oct 2026.

In brief

Draft created by Release watch. Claims have not been checked against independent data yet. Review each claim against the source before publishing.

Claims examined

- None

Models and list prices

Model	Input	Output
claude-sonnet-4-5 claude-sonnet-4-5	–	–
claude-sonnet-4-5-20250929 claude-sonnet-4-5-20250929	–	–

$ per 1M tokens.

So what

There is no independent evidence yet. Run the live pack before relying on these numbers.

Exhibit 21

What independent data shows

gemini-1.5-flash-latest against gemini-1.5-flash

Released 1 Oct 2026. Sources checked 1 Oct 2026.

In brief

Draft created by Release watch. Claims have not been checked against independent data yet. Review each claim against the source before publishing.

Claims examined

- None

Models and list prices

Model	Input	Output
gemini-1.5-flash gemini-1.5-flash	–	–
gemini-1.5-flash-latest gemini-1.5-flash-latest	–	–

$ per 1M tokens.

So what

There is no independent evidence yet. Run the live pack before relying on these numbers.

Exhibit 22

What independent data shows

gemini-1.5-pro-latest against gemini-1.5-pro

Released 1 Oct 2026. Sources checked 1 Oct 2026.

In brief

Draft created by Release watch. Claims have not been checked against independent data yet. Review each claim against the source before publishing.

Claims examined

- None

Models and list prices

Model	Input	Output
gemini-1.5-pro gemini-1.5-pro	–	–
gemini-1.5-pro-latest gemini-1.5-pro-latest	–	–

$ per 1M tokens.

So what

There is no independent evidence yet. Run the live pack before relying on these numbers.

Exhibit 23

What independent data shows

Gemini 4 Argon against Claude Opus 5.5

Released 30 Sept 2026. Sources checked 1 Oct 2026.

In brief

Google's new frontier model leads its chosen rivals on the Vals Index and AutomationBench, and independent leaderboards confirm both. Its headline price of $2/$10 is introductory and doubles later; at the later price its cost per AutomationBench task is above Opus 5.5. Argon is open only to selected cyber partners, so no live test is possible yet.

Claims examined

- Leads rival frontier models on the Vals Index, ahead of Opus 5.5[19, 20]
- Ranks first on Zapier's AutomationBench at 51.3% [19, 20]
- Finds far more source-code vulnerabilities than Gemini 3.8 Flash Cyber (85.8% vs 71.0%)[19, 25]

Our additional checks

- Half Opus 5.5's token price should mean about half the cost per task (hypothesis)
- The same saving, after the introductory price ends (stress test)
- The saving on vulnerability-fixing work (CWE-bench) (stress test)

Models and list prices

Model	Input	Output
Claude Opus 5.5 claude-opus-5-5	$4.00	$20.00
Gemini 4 Argon (introductory price) gemini-4-argon	$2.00	$10.00
Gemini 4 Argon (later list price) gemini-4-argon	$4.00	$20.00
Gemini 3.8 Flash Cyber gemini-3.8-flash-cyber	$0.75	$3.75

$ per 1M tokens. Source:[19]

So what

The claim holds under the vendor's stated conditions but not outside them. Pin the setting you deploy in your contract or configuration, and test at that setting.

Exhibit 24

What independent data shows

Hallmarked
100% of claim

Leads rival frontier models on the Vals Index, ahead of Opus 5.5[19, 20]

Google. Claimed: 1.9 points higher. Independent data: 1.9 points higher.



Published figures: 67 → 68.9 Vals Index, %. Vals' own leaderboard, which publicly named Argon its new #1. The lead is 1.9 points.[21, 22]

Cleared for reuse until 30 Dec 2026. Conditions ID f8d7c63e.

Hallmarked
100% of claim

Ranks first on Zapier's AutomationBench at 51.3%[19, 20]

Google. Claimed: 8.8 points higher. Independent data: 8.8 points higher.



Published figures: 42.47 → 51.29 AutomationBench, % tasks fully completed. Zapier's live leaderboard: Argon High against Opus 5.5 Max with default fallbacks.[23, 22]

Cleared for reuse until 30 Dec 2026. Conditions ID 6d62c86f.

Hallmarked
82% of claim

Half Opus 5.5's token price should mean about half the cost per task

Assay hypothesis

Assay, from Google's $2/$10 price. Claimed: 50% cheaper. Independent data: 41% cheaper.



Published figures: $1.44 → $0.85 per AutomationBench task. Zapier's reported cost per task at Argon's promotional pricing.[23, 22] Google states the price, not a per-task saving. This row tests what the price implies, at the introductory rate.

Cleared for reuse until 30 Dec 2026. Conditions ID d657f2eb.

Reversed
outside stated terms

The same saving, after the introductory price ends

Assay stress test

Assay, from Google's $2/$10 price. Claimed: 50% cheaper. Independent data: 18% more expensive.



Published figures: $1.44 → $1.7 per AutomationBench task. Zapier's reported cost per task at Argon's standard list price.[23, 22] Google's footnote says prices double to $4/$20 later. This row shows the per-task cost at that list price.

Cleared for reuse until 30 Dec 2026. Conditions ID b0cc360e.

Reversed
outside stated terms

The saving on vulnerability-fixing work (CWE-bench)

Assay stress test

Assay, from Google's $2/$10 price. Claimed: 50% cheaper. Independent data: 739% more expensive.



Published figures: $0.79 → $6.63 per CWE-bench rollout. Evaluator's own cost accounting; Argon ran in Antigravity, Opus in Claude Code.[24, 22] A different workload. Scores are close (68% vs 67%) but each model ran in its own agent harness.

Cleared for reuse until 30 Dec 2026. Conditions ID 0393a90a.

Unassayed
no public data

Finds far more source-code vulnerabilities than Gemini 3.8 Flash Cyber (85.8% vs 71.0%)[19, 25]

Google. Claimed: 15 points higher.
Google's internal evaluation only. Argon is open to selected Fairwind partners, so no one outside Google has published a replication yet.

What else the data shows

- Argon is not generally available: access starts with selected cyber-defence partners, and Google has not published a public API model ID, so live tests are not possible yet.[19, 22]
- Coding results are mixed: Argon leads DeepSWE v1.1 but trails GPT-6 Astra on FrontierSWE v2 (55.0% vs 65.5%) and Opus 5.5 on Terminal-bench 4.0 (57.4% vs 66.4%).[20, 22]
- Google's comparison leaves out Claude Sonnet 5.5, which scores higher on Vibe Code Bench, and GPT-6.1 Sol, which already sells at the same $2/$10 price.[22]
- The headline 1 million tokens is an output limit, not input context; Google has not stated the commercial input limit.[19, 22]
- Google ran Argon at its highest thinking setting, and some rows use unequal conditions, such as different video frame budgets on LVBench.[25, 22]

Claude Sonnet 5.5 against Sonnet 5

Released 28 Sept 2026. Sources checked 30 Sept 2026.

In brief

Anthropic kept Sonnet 5's $2/$10 price and says Sonnet 5.5 costs up to 30% less per task and writes output more than 30% faster. Launch-day independent data is at max effort only, and there the saving reverses: Sonnet 5.5 writes far more tokens and costs more per task than Sonnet 5, while scoring much higher. The vendor claims are about lower effort, which the live pack tests.

Claims examined

- Costs up to 30% less per task than Sonnet 5[26, 27]
- Generates output more than 30% faster than Sonnet 5[26, 27]
- Uses fewer output tokens per task (Slack reported about 14% fewer)[26, 29]

Our additional checks

- The same per-task saving, checked at max effort (stress test)
- Fewer output tokens per task, checked at max effort (stress test)

Models and list prices

Model	Input	Output
Sonnet 5 (default) claude-sonnet-5	$2.00	$10.00
Sonnet 5.5 (medium) claude-sonnet-5-5	$2.00	$10.00
Sonnet 5 (max) claude-sonnet-5	$2.00	$10.00
Sonnet 5.5 (max) claude-sonnet-5-5	$2.00	$10.00

$ per 1M tokens. Source:[26]

So what

The claim holds under the vendor's stated conditions but not outside them. Pin the setting you deploy in your contract or configuration, and test at that setting.

Exhibit 25

What independent data shows

Unassayed
no public data

Costs up to 30% less per task than Sonnet 5[26, 27]

Anthropic. Claimed: 30% cheaper.

Launch-day independent figures are at max effort only (see the stress test). The live pack runs Sonnet 5.5 at medium effort against Sonnet 5 at its default. Anthropic ties its efficiency claims to lower effort settings.

Reversed
outside stated terms

The same per-task saving, checked at max effort Assay stress test

Anthropic. Claimed: 30% cheaper. Independent data: 49% more expensive.



Published figures: $5.09 → $7.6 per index task. Artificial Analysis Intelligence Index v4.3.2, both at max effort, as reported on launch day.[28, 29] Outside the conditions Anthropic stated. This row shows what happens if you run both at max.

Review before reuse. A newer model in the same family appeared after verification: claude-sonnet-4-5-20250929. Conditions ID 01afd1b4.

Unassayed
no public data

Generates output more than 30% faster than Sonnet 5[26, 27]

Anthropic. Claimed: 30% faster.

No independent output-speed figure had been published at launch. Re-check when Artificial Analysis adds one; the live pack measures latency meanwhile.

Unassayed
no public data

Uses fewer output tokens per task (Slack reported about 14% fewer)[26, 29]

Anthropic, quoting customers. Claimed: 14% fewer tokens.

Customer figures come from each customer's own evaluation at production settings. No independent figure at medium effort yet; run the live pack. Tested against the Slack figure, one of the lower customer numbers quoted.

Reversed
outside stated terms

Fewer output tokens per task, checked at max effort Assay stress test

Anthropic, quoting customers. Claimed: 14% fewer tokens. Independent data: 64% more tokens.



Published figures: 118 → 193 thousand output tokens per task. Roughly 1.6 times the output tokens per task, most of it reasoning.[28, 29] The token claim is not stated for max effort. This row shows the setting where it breaks.

Review before reuse. A newer model in the same family appeared after verification: claude-sonnet-4-5-20250929. Conditions ID a1b8557c.

What else the data shows

- List prices did not change ($2/$10, cache reads $0.20), so any per-task saving must come from using fewer tokens, not from price.[26, 27]
- Default effort on the API is high, while the Claude apps default to medium; Anthropic advises re-running effort sweeps rather than carrying Sonnet 5 settings over.[30, 29]
- Five API changes can break Sonnet 5 code, including thinking that can no longer be disabled, removal of forced tool choice, and errors on non-default sampling parameters.[30, 29]
- Independent Terminal-Bench 4.0 came in at 64% against Anthropic's 70.6%; Artificial Analysis also scores it lower than Opus 5.5 on its knowledge-versus-hallucination index.[29]

Claude Opus 5.5 against Opus 5

Released 22 Sept 2026. Sources checked 25 Sept 2026.

In brief

Anthropic cut Opus prices by 20% and says Opus 5.5 costs about 40% less to run than Opus 5 at default effort, while generating output more than 30% faster. Independent measurements support the claim at the default medium effort. At max effort the story reverses: Opus 5.5 writes far more tokens and costs about the same per task.

Claims examined

- Costs about 40% less to run than Opus 5 on typical workloads at default effort[32]
- Generates output more than 30% faster than Opus 5[32]
- Uses fewer tokens per task (customer reports range from 20% to 66% fewer)[32]

Our additional checks

- The same 40% saving, checked at max effort (stress test)
- Fewer tokens per task, checked at max effort (stress test)

Models and list prices

Model	Input	Output
Opus 5 (medium) claude-opus-5	$5.00	$25.00
Opus 5.5 (medium) claude-opus-5-5	$4.00	$20.00
Opus 5 (max) claude-opus-5	$5.00	$25.00
Opus 5.5 (max) claude-opus-5-5	$4.00	$20.00

$ per 1M tokens. Source:[32]

So what

The claim holds under the vendor's stated conditions but not outside them. Pin the setting you deploy in your contract or configuration, and test at that setting.

Exhibit 26

What independent data shows

Hallmarked
97% of claim

Costs about 40% less to run than Opus 5 on typical workloads at default effort[32]

Anthropic. Claimed: 40% cheaper. Independent data: 39% cheaper.



Published figures: $2.19 → $1.34 per index task. Artificial Analysis Intelligence Index v4.3.2, both at medium effort.[33, 34]

Review before reuse. A newer model in the same family appeared after verification: claude-opus-4-8. Conditions ID a02ad754.

Reversed
outside stated terms

The same 40% saving, checked at max effort Assay stress test

Anthropic. Claimed: 40% cheaper. Independent data: 2% more expensive.



Published figures: $5.86 → $5.98 per index task. Artificial Analysis Intelligence Index v4.3.2, both at max effort.[35, 36] Anthropic's claim is for default effort. This row shows what happens if you run both at max.

Review before reuse. A newer model in the same family appeared after verification: claude-opus-4-8. Conditions ID 2c56a953.

Hallmarked
121% of claim

Generates output more than 30% faster than Opus 5[32]

Anthropic. Claimed: 30% faster. Independent data: 36% faster.



Published figures: 55 → 75 output tokens per second. Measured on Anthropic's API at medium effort.[33, 34]

Review before reuse. A newer model in the same family appeared after verification: claude-opus-4-8. Conditions ID 62954a23.

Hallmarked
112% of claim

Uses fewer tokens per task (customer reports range from 20% to 66% fewer)[32]

Anthropic, quoting customers. Claimed: 20% fewer tokens. Independent data: 22% fewer tokens.



Published figures: 49 → 38 million tokens across the index. Total tokens generated on the index at medium effort.[33, 34] Tested against the lowest customer figure, 20%.

Review before reuse. A newer model in the same family appeared after verification: claude-opus-4-8. Conditions ID f5f717b0.

Reversed
outside stated terms

Fewer tokens per task, checked at max effort Assay stress test

Anthropic, quoting customers. Claimed: 20% fewer tokens. Independent data: 63% more tokens.



Published figures: 73 → 119 thousand output tokens per task. Roughly 1.6 times the output tokens per task.[37] The token claim is not specific to max effort. This row shows the setting where it breaks.

Review before reuse. A newer model in the same family appeared after verification: claude-opus-4-8. Conditions ID 2814889e.

What else the data shows

- Per-token prices fell 20% ($5/$25 to $4/$20) and cache reads fell 60% ($0.50 to $0.20), so part of the saving is pricing, not efficiency.[32, 37]
- Thinking can no longer be switched off on Opus 5.5, so pipelines that ran Opus 5 without thinking lose that cheaper option.[32]

GPT-6 Sol and Luna against GPT-5.6

Released 22 Sept 2026. Sources checked 25 Sept 2026.

In brief

OpenAI halved token prices for its mid and low tiers. Independent measurement confirms cost per task roughly halves, driven entirely by price: both models use slightly more output tokens than before, and Luna's benchmark score dropped two points.

Claims examined

- GPT-6 Sol costs half as much as GPT-5.6 Sol[39]
- GPT-6 Luna cuts prices by about half[39]
- Sol makes about half as many factual mistakes as GPT-5.6 Sol[39]

Models and list prices

Model	Input	Output
GPT-5.6 Sol gpt-5.6-sol	$4.00	$20.00
GPT-6 Sol gpt-6-sol	$2.00	$10.00
GPT-5.6 Luna gpt-5.6-luna	$0.20	$1.20
GPT-6 Luna gpt-6-luna	$0.10	$0.50

$ per 1M tokens. Source:[39]

So what

The saving is real, but part of it came with lower measured quality. Check your own quality bar before switching.

Exhibit 27

What independent data shows

Hallmarked
93% of claim

GPT-6 Sol costs half as much as GPT-5.6 Sol[39]

OpenAI. Claimed: 50% cheaper. Independent data: 47% cheaper.



Published figures: $1.99 → $1.06 per index task. Artificial Analysis Intelligence Index v4.3.2, both at max effort.[40] OpenAI states the cut per token; this checks it per completed task.

Cleared for reuse until 24 Dec 2026. Conditions ID 04f9f766.

Plated
122% of claim

GPT-6 Luna cuts prices by about half[39]

OpenAI. Claimed: 50% cheaper. Independent data: 61% cheaper.



Published figures: $0.18 → $0.07 per index task. Artificial Analysis Intelligence Index v4.3.2, both at max effort.[40]

Intelligence Index fell from 43 to 41, so the gain came with lower measured quality.

Cleared for reuse until 24 Dec 2026. Conditions ID c921487f.

Unassayed
no public data

Sol makes about half as many factual mistakes as GPT-5.6 Sol[39]

OpenAI. No figure that can be tested directly.

OpenAI's figure comes from its own set of user-flagged conversations. Artificial Analysis reports its knowledge and hallucination index rising from 22 to 27, which points the same way but measures something different. Test it with a factual task set of your own.

What else the data shows

- The saving comes from price: output tokens per task rose slightly, from about 29k to 31k for Sol and 41k to 51k for Luna.[41]
- Independent evals show regressions too: Sol dropped about 100 Elo on GDPval-AA v2.1, a benchmark of economically valuable tasks.[41]
- Prompts above 272K input tokens are billed at higher long-context rates, so very long-context workloads save less.[42]

Gemini 3.8 Flash against 3.7 Flash

Released 2 Sept 2026. Sources checked 25 Sept 2026.

In brief

Same price, same limits, three weeks apart. Google leads with a customer saying 3.8 Flash completed more than three times as many tasks. Independent indexes show a real but much smaller gain in agentic work, flat coding, and a slower first token.

Claims examined

- Completed more than three times as many tasks as 3.7 Flash[46]
- Agentic video understanding uses up to 88% fewer tokens on long-form content[45]

Models and list prices

Model	Input	Output
Gemini 3.7 Flash gemini-3.7-flash	$0.75	$3.75
Gemini 3.8 Flash gemini-3.8-flash	$0.75	$3.75

$ per 1M tokens. Source:[45]

So what

The headline figure comes from a single customer's evaluation; independent data shows a much smaller gain. Treat it as an upper bound and measure on your own workload.

Exhibit 28

What independent data shows

Fool's gold
5% of claim

Completed more than three times as many tasks as 3.7 Flash[46] Proxy measure

Customer quote on Google's Flash page. Claimed: 200% higher score. Independent data: 11% higher score.



Published figures: 45.1 → 50 Agentic Index points. Artificial Analysis Agentic Index.[46] The quote comes from one company's document-heavy evaluation. The reference is an independent agentic index, so it is a proxy, not a replication.

Cleared for reuse until 24 Dec 2026. Conditions ID a2fcfdc6.

Unassayed
no public data

Agentic video understanding uses up to 88% fewer tokens on long-form content[45]

Google. Claimed: 88% fewer tokens.
No independent measurement found. Build a long-video task set and run it live; note the claim says up to 88%, so a typical figure will be lower.

What else the data shows

- Google itself cautions that the model may use more tokens to maximise performance, especially at higher effort levels, so cost per task can rise even at an unchanged price.[47]
- Time to first token got worse, from 12.0 to 13.2 seconds, even though streaming speed rose from 279 to 299 tokens per second.[46]

- The independent coding index barely moved (76.1 to 76.3), while the overall Intelligence Index rose from 56 to 59.[46]
- Both models are on introductory pricing of $0.75/$3.75 that rises to $1.50/$7.50 on 1 January 2027.[48]

Claude Opus 4.5 token efficiency against Sonnet 4.5

Released 24 Nov 2025. Sources checked 25 Sept 2026.

In brief

A widely repeated efficiency claim from late 2025 that was measured on one benchmark under the vendor's own harness. There is no matching independent number, so this scenario is a live test pack only.

Claims examined

- At medium effort, matches Sonnet 4.5 on SWE-bench Verified with 76% fewer output tokens[49]

Our additional checks

- Cheaper per task than Sonnet 4.5 despite higher token prices (hypothesis)

Models and list prices

Model	Input	Output
Sonnet 4.5 claude-sonnet-4-5	$3.00	$15.00
Opus 4.5 (medium) claude-opus-4-5-20251101	$5.00	$25.00

$ per 1M tokens. Source:[49]

> **So what**
> There is no independent evidence yet. Run the live pack before relying on these numbers.

Exhibit 29

What independent data shows

Unassayed
no public data

At medium effort, matches Sonnet 4.5 on SWE-bench Verified with 76% fewer output tokens[49]

Anthropic. Claimed: 76% fewer tokens.
The figure is from one coding benchmark on Anthropic's harness. The live pack tests whether the saving carries over to short, general tasks, where it is likely to be smaller.

Unassayed
no public data

Cheaper per task than Sonnet 4.5 despite higher token prices Assay hypothesis

Implied by the token and price claims. Claimed: 20% cheaper.
Opus 4.5 costs $5/$25 against Sonnet 4.5's $3/$15, so it only wins per task if it really writes far fewer tokens. The 20% target here is illustrative.

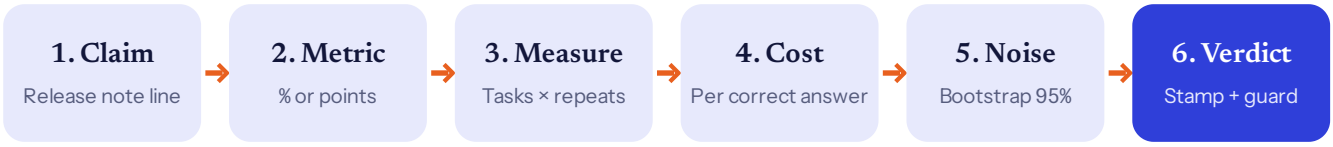
What else the data shows

- At high effort, Anthropic says Opus 4.5 beat Sonnet 4.5 by 4.3 points while using 48% fewer tokens, so the saving shrinks as effort rises.[49]

Every verdict follows the same six steps, and every number traces back to a source.

Exhibit 30

The Assay validation pipeline



Rule, applied in order	Verdict
Live runs only: the 95% bootstrap range crosses zero	Unassayed
The measure moved the wrong way	Fool's gold
80% or more of the claimed improvement delivered	Hallmarked
40% to 79% delivered, or 80%+ delivered while accuracy or the benchmark index fell by 2 points or more	Plated
Under 40% delivered	Fool's gold
No independent measurement exists	Unassayed

Improvements are relative for cost, tokens, latency and speed, and absolute points for accuracy and benchmark indexes. Cost per task counts input tokens and all output tokens, including hidden reasoning, at list price. Vendor claims are tested as stated; stress tests re-run a claim outside the vendor's stated conditions and are reported separately. Live runs grade every answer automatically, repeat each task, and use a paired bootstrap over tasks with 800 resamples. Reference verdicts compare two published figures and carry no noise range.

Limitations

Independent benchmarks use their own task mixes, harnesses and effort settings, which may not match yours. Where no primary figure was published, secondary reporting is cited and labelled as such. Model IDs marked for confirmation, and draft scenarios extracted automatically from release notes, should be reviewed before decisions are made. This report was generated by Assay on 13 Nov 2026.

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Vendor: the company's own announcement or pricing page. Independent: a third party that ran its own measurements. Secondary: reporting that summarises either.

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